

INSTITUTO SUPERIOR DE ENGENHARIA DO PORTO

Pós-Graduação em Sistemas Computacionais Embebidos
Sistemas Operativos de Tempo-Real (PGSCE-SIOPTR)

SIOPTR: TT3: Week 5

Linux Kernel Modules (cont.)

Paulo Baltarejo Sousa
pbs@isep.ipp.pt
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1 Learning examples

Download the learning examples from <https://drive.google.com/file/d/1S69yq-9DUt-7ak57FRjKUpZGaeXvDeHf/view?usp=sharing>. Download LKM2.tar.gz compressed file. In this compressed file you can find several source code modules and a Makefile. For extracting it type:

```
> tar -xvzf LKM2.tar.gz
```

1.1 kthread.c

This module shows a basic implementation of a two kernel threads. Each one executes periodically.

1. Compilation:

```
> make FILE=kthread
```
2. Insert it into the Linux kernel:

```
> sudo insmod ./kthread.ko
```
3. Check printk messages:

```
> sudo dmesg | grep "LKM"
```
4. Remove it from the Linux kernel:

```
> sudo rmmod kthread
```

1.2 kthread2.c

This module shows a basic implementation of a two kernel threads. One of the threads is activated every second and its only task is to activate the second one, which prints messages. Whenever the second one is not performing its task (printing messages), it should wait in a waitqueue.

1. Compilation:

```
> make FILE=kthread2
```
2. Insert it into the Linux kernel:

```
> sudo insmod ./kthread2.ko
```
3. Check printk messages:

```
> sudo dmesg | grep "LKM"
```
4. Remove it from the Linux kernel:

```
> sudo rmmod kthread2
```

1.3 work.c

This module shows a basic implementation of a workqueue. It adds two tasks to the workqueue.

1. Compilation: `> make FILE=work`
2. Insert it into the Linux kernel:
`> sudo insmod ./work.ko`
3. Check printk messages:
`> sudo dmesg | grep "LKM"`
4. Remove it from the Linux kernel:
`> sudo rmmmod work`

1.4 timer.c

This module shows an implementation of a low resolution timer. It counts the number of seconds from its insertion

1. Compilation: `> make FILE=timer`
2. Insert it into the Linux kernel:
`> sudo insmod ./timer.ko`
3. Change /proc/timer permissions:
`> sudo chmod 666 /proc/timer`
4. Open a console and type:
`> cat /proc/timer`
5. Check printk messages:
`> sudo dmesg | grep "LKM"`
6. Remove it from the Linux kernel:
`> sudo rmmmod timer`

1.5 hrtimer.c

This module shows an implementation of a high resolution timer. It counts the number of seconds from its insertion

1. Compilation: `> make FILE=hrtimer`

2. Insert it into the Linux kernel:
 > `sudo insmod ./hrtimer.ko`
3. Change /proc/timer permissions:
 > `sudo chmod 666 /proc/hrtimer`
4. Open a console and type:
 > `cat /proc/hrtimer`
5. Check printk messages:
 > `sudo dmesg | grep "LKM"`
6. Remove it from the Linux kernel:
 > `sudo rmmod hrtimer`

1.6 hrtimer2.c

This module presents the values of different timer sources

1. Compilation: > `make FILE=hrtimer2`
2. Insert it into the Linux kernel:
 > `sudo insmod ./hrtimer2.ko`
3. Check printk messages:
 > `sudo dmesg | grep "LKM"`
4. Remove it from the Linux kernel:
 > `sudo rmmod hrtimer2`